Lecture 9

The canonical pair: Weyl's relation and Stone–von Neumann

Objectives. *We prove the Stone–von Neumann theorem: every irreducible Weyl pair of strongly continuous unitary groups is unitarily equivalent to the pair on the line. The proof builds a nonzero projection from Gaussian integrals. The box shows why the commutator on a dense domain is insufficient.*

- Derive Weyl's relation from translations and phase multiplication.
- Construct the unitary equivalence from the Gaussian projection.
- Explain the roles of strong continuity, Weyl's relation and irreducibility.

9.1 Translations and Weyl's relation

On the line the momentum moves states. We write $U(a)$, with its argument in parentheses, for a translation by the distance a, and keep U_t, with a subscript, for the evolution.

Proposition 9.1 (Momentum generates translations). *For real a and $\psi \in L^2(\mathbb{R})$, put $(U(a)\psi)(x) = \psi(x-a)$. Then $(U(a))_{a\in\mathbb{R}}$ is a strongly continuous unitary group with the generator $P/\hbar$. So $U(a) = e^{-\mathrm{i}aP/\hbar}$, and $a \mapsto U(a)\psi$ is differentiable exactly when $\psi \in D(P)$.*

Proof. The substitution $y = x - a$ shows that $U(a)$ keeps norms, and $U(-a)$ is its inverse, so $U(a)$ is unitary. By the self-check of § 4.1, the transform of $U(a)\psi$ is $e^{-\mathrm{i}pa/\hbar}\tilde{\psi}$ for $\psi \in \mathcal{S}(\mathbb{R})$. So $U(a)$ agrees with $F^{-1}M_{e^{-\mathrm{i}ap/\hbar}}F$ on $\mathcal{S}(\mathbb{R})$, which is dense by the standing fact S4. Both are bounded, so they agree on every vector, as at the end of the proof of Proposition 7.9. By the multiplication case of Theorem 7.4, with the function p and the unitary F, this is a strongly continuous unitary group, and its generator is $F^{-1}M_pF/\hbar = P/\hbar$. The differentiability claim is Proposition 7.3. ∎

So $D(P)$ is the set of states whose translations are differentiable in a, and for $\psi \in \mathcal{S}(\mathbb{R})$ the generator acts as $\mathrm{i}\,\mathrm{d}\psi(x-a)/\mathrm{d}a$ at $a = 0$, which is $-\mathrm{i}\psi' = P\psi/\hbar$.

The position generates a second group. For real b, let $V(b)$ multiply by the phase $e^{\mathrm{i}bx/\hbar}$. We keep its argument, since V alone will be a potential from Lecture 10 on. By the multiplication case of Theorem 7.4, with the function $-x$ and the unitary $\mathbb{1}$, the operators $V(b)$ form a strongly continuous unitary group with the generator $-Q/\hbar$, so $V(b) = e^{\mathrm{i}bQ/\hbar}$. The translation $U(a)$ moves the position by a, and $V(b)$ moves the momentum by b. Indeed, by the self-check of § 4.1, the transform of $V(b)\psi$ is $\tilde{\psi}(p-b)$ for $\psi \in \mathcal{S}(\mathbb{R})$.

Proposition 9.2 (Weyl's relation). *For all real a and b,*

$$U(a)V(b) = e^{-\mathrm{i}ab/\hbar}\,V(b)U(a).$$

Proof. For $\psi \in L^2(\mathbb{R})$, $(U(a)V(b)\psi)(x) = e^{\mathrm{i}b(x-a)/\hbar}\psi(x-a)$, and $(V(b)U(a)\psi)(x) = e^{\mathrm{i}bx/\hbar}\psi(x-a)$. So the first is $e^{-\mathrm{i}ab/\hbar}$ times the second. ∎

In *Quantum Mechanics I*, Thm 9.2, the clock and the shift on $\mathbb{C}^N$ obey $DT = \zeta TD$. That lecture promised the form of Weyl's relation that this course proves, and Proposition 9.2 is that form. Both translations together move a state through phase space. Interchanging their order multiplies the vector by $e^{-\mathrm{i}ab/\hbar}$.

Definition 9.3 (Weyl pair)**.** A *Weyl pair* on a Hilbert space $\mathcal{H} \neq \{0\}$ is two strongly continuous unitary groups $(U'(a))_{a\in\mathbb{R}}$ and $(V'(b))_{b\in\mathbb{R}}$ on $\mathcal{H}$ with

$$U'(a)V'(b) = e^{-\mathrm{i}ab/\hbar}\,V'(b)U'(a) \quad \text{for all real } a \text{ and } b.$$

Its *Weyl operators* are $W'(q,p) = e^{-\mathrm{i}qp/2\hbar}V'(p)U'(q)$. The pair is *irreducible* if the only closed subspaces of $\mathcal{H}$ that every $U'(a)$ and every $V'(b)$ map into themselves are $\{0\}$ and $\mathcal{H}$.

By Propositions 9.1 and 9.2 and the paragraph between them, the groups $U(a)$ and $V(b)$ of the line form a Weyl pair, the *Schrödinger pair*, with the Weyl operators $W(q,p) = e^{-\mathrm{i}qp/2\hbar}V(p)U(q)$, that is,

$$(W(q,p)\psi)(x) = e^{-\mathrm{i}qp/2\hbar}\,e^{\mathrm{i}px/\hbar}\,\psi(x-q).$$

A prime marks the operators of a general pair. We write $z = (q,p)$ for a point of phase space, and

$$\sigma(z,w) = q'p - qp' \qquad \text{for } z = (q,p) \text{ and } w = (q',p').$$

The phase $\sigma(z,w)$ is linear in z and in w, and $\sigma(w,z) = -\sigma(z,w)$.

Proposition 9.4 (The product law)**.** *Let $(U'(a))$ and $(V'(b))$ be a Weyl pair. Each $W'(z)$ is unitary, $z \mapsto W'(z)\psi$ is continuous for every $\psi \in \mathcal{H}$, and for all points z and w of phase space*

$$W'(z)W'(w) = e^{\mathrm{i}\sigma(z,w)/2\hbar}\,W'(z+w).$$

In particular, $W'(z)^\dagger = W'(-z)$, $W'(q,0) = U'(q)$ and $W'(0,p) = V'(p)$.

Proof. A product of unitaries is unitary. Let $z_n \to z$, with $z_n = (q_n, p_n)$. Then

$$\|V'(p_n)U'(q_n)\psi - V'(p)U'(q)\psi\| \le \|U'(q_n)\psi - U'(q)\psi\| + \|V'(p_n)U'(q)\psi - V'(p)U'(q)\psi\|,$$

since $V'(p_n)$ keeps norms, and both terms tend to 0. For the product, Weyl's relation with $a = q$ and $b = p'$ moves $U'(q)$ past $V'(p')$:

$$V'(p)U'(q)V'(p')U'(q') = e^{-\mathrm{i}qp'/\hbar}\,V'(p+p')U'(q+q').$$

With the phases of the three Weyl operators, the exponent is $\mathrm{i}/\hbar$ times

$$-\frac{qp+q'p'}{2} - qp' + \frac{(q+q')(p+p')}{2} = \frac{q'p-qp'}{2} = \frac{\sigma(z,w)}{2}.$$

Finally, $\sigma(z,-z) = 0$ gives $W'(z)W'(-z) = W'(0) = \mathbb{1}$, so $W'(z)^\dagger = W'(z)^{-1} = W'(-z)$. ∎

For each z, the operators $W'(tz)$ form a unitary group in t, since $\sigma(tz,t'z) = 0$. For $\psi \in \mathcal{S}(\mathbb{R})$, the derivative in t of $(W(tq,tp)\psi)(x)$ at $t=0$ is $(\mathrm{i}/\hbar)(px\psi(x) - q(P\psi)(x))$. This is why $W(q,p)$ is often written $e^{\mathrm{i}(pQ-qP)/\hbar}$. We use only the definition.

One matrix element will matter. Let $s > 0$ be a length. Every s is the oscillator length of Lecture 8 for the frequency $\nu = \hbar/ms^2$, and it is not the width s of Lecture 4. With $p_0 = p$, Theorem 8.10 says that $W(q,p)h_0$ is the coherent state ψ_α with $\alpha = (q/s + \mathrm{i}sp/\hbar)/\sqrt{2}$, where h_0 is the ground state $h_0(x) = (\pi s^2)^{-1/4}e^{-x^2/2s^2}$ of an oscillator with the length s (Lecture 8). By its series and the orthonormality of the h_n, $\langle h_0, \psi_\alpha\rangle = e^{-|\alpha|^2/2}$. So, with $g(q,p) = q^2/s^2 + s^2p^2/\hbar^2$,

$$\langle h_0, W(z)h_0\rangle = e^{-g(z)/4} \quad \text{for every } z. \tag{9.1}$$

Self-check 9.1

1. Let $\psi \in \mathcal{S}(\mathbb{R})$ be a unit vector. Show that the mean position in $U(a)\psi$ is $\langle Q\rangle + a$.

9.2 The theorem, and why $[Q,P] = \mathrm{i}\hbar$ is not enough

Theorem 9.5 (Stone–von Neumann). *Let $(U'(a))$ and $(V'(b))$ be an irreducible Weyl pair on $\mathcal{H}$. Then there is a unitary T from $\mathcal{H}$ onto $L^2(\mathbb{R})$ with*

$$TU'(a) = U(a)T \quad \text{and} \quad TV'(b) = V(b)T \quad \text{for all real } a \text{ and } b.$$

§ 9.4 proves it. The unitary T carries the generators along. If $(U'(h)\psi - \psi)/h$ converges, then so does $(U(h)T\psi - T\psi)/h = T(U'(h)\psi - \psi)/h$, and conversely, by T^{-1}. So T maps the domain of the generator of $(U'(a))$ onto $D(P)$, and that generator to $P/\hbar$ (Definition 7.2). In the same way it carries the generator of $(V'(b))$ to $-Q/\hbar$. In words, an irreducible Weyl pair is the position and the momentum of a particle on the line, with their domains, up to a unitary change of the state space. *Quantum Mechanics I*, Exercise Sheet 9, proves the finite form: two unitaries on $\mathbb{C}^N$ that obey the relations of the clock and the shift, Weyl's relation and N-th powers equal to $\mathbb{1}$, are the clock and the shift in another basis.

The groups contain information that the commutator does not. The relation $QP - PQ = \mathrm{i}\hbar$ also holds in a box, and there it decides nothing. In a box, a translation must say what happens at the walls. For $\theta \in [0, 2\pi)$, extend $\psi \in L^2(0,L)$ to the line by $\psi(x+L) = e^{\mathrm{i}\theta}\psi(x)$, and let $(U_\theta(a)\psi)(x) = \psi(x-a)$ for $0 < x < L$. What leaves through the right wall enters at the left, multiplied by $e^{-\mathrm{i}\theta}$. Since $|\psi|$ extends with the period L, $U_\theta(a)$ keeps norms. The eigenvector $e_{\theta,k}$ of Theorem 3.10 is its own extension, and $e_{\theta,k}(x-a) = e^{-\mathrm{i}a(\theta+2\pi k)/L}e_{\theta,k}(x)$. So $U_\theta(a)$ agrees on the basis $(e_{\theta,k})$, complete by the standing fact S2, with the group of the diagonal operator with these eigenvectors and the eigenvalues $\hbar(\theta+2\pi k)/L$, which is unitary by Theorem 7.4. Both are linear, so they agree on the finite combinations of the basis, which are dense. Both keep norms, so they agree on every vector, as at the end of the proof of Proposition 7.9. Hence $(U_\theta(a))$ is a strongly continuous unitary group, and by Theorem 7.4 its generator is that diagonal operator divided by $\hbar$. By the sketched Theorem 3.10, the diagonal operator is P_θ.

In the box, Q multiplies by x, and it is bounded, with $\|Q\psi\| \le L\,\|\psi\|$. Let $V(b)$ multiply by $e^{\mathrm{i}bx/\hbar}$ on $L^2(0,L)$. Since $|x| \le L$ there, the bounds (7.2) give $(V(b)\psi - \psi)/b \to \mathrm{i}Q\psi/\hbar$ in norm for every ψ, so $(V(b))$ is a strongly continuous unitary group with the generator $-Q/\hbar$.

The smooth functions that vanish near both walls form a dense subspace $C_c^\infty(0,L)$ (S4). The operators Q and P_θ map it into itself, and on it $QP_\theta\psi - P_\theta Q\psi = \mathrm{i}\hbar\psi$, by the product rule. Yet Weyl's relation fails. By the definition of the extension,

$$(U_\theta(L)\psi)(x) = \psi(x-L) = e^{-\mathrm{i}\theta}\psi(x),$$

so $U_\theta(L) = e^{-\mathrm{i}\theta}\mathbb{1}$. Weyl's relation at $a = L$ would then read $e^{-\mathrm{i}\theta}V(b) = e^{-\mathrm{i}Lb/\hbar}e^{-\mathrm{i}\theta}V(b)$, that is, $e^{-\mathrm{i}Lb/\hbar} = 1$ for every b, which is false for $b = \pi\hbar/L$. Nor is the box's pair unitarily equivalent to the line's. A unitary from $L^2(0, L)$ onto $L^2(\mathbb{R})$ that carried the box's Q to the line's would make the line's Q bounded and defined on every vector, and it is not. The function $(1+|x|)^{-1}$ is square-integrable, but it lies outside its domain.

So the commutator on a dense domain does not fix the pair. The theorem needs the groups. Hall, Ex. 14.5, makes the same point for $\theta = 0$ on the interval $[-1, 1]$.

Self-check 9.2

1. Show that $U_\theta(L) = e^{-\mathrm{i}\theta}\mathbb{1}$ from the eigenvalues of P_θ.

9.3 Von Neumann's projection

On the line, the projection $|h_0\rangle\langle h_0|$ onto the oscillator's ground state picks one direction out of $L^2(\mathbb{R})$, and the Weyl operators carry h_0 to vectors whose closed span is $L^2(\mathbb{R})$ (Proposition 8.11). Von Neumann's idea is to build this projection from the Weyl operators alone, so that it exists for every Weyl pair. Fix the length $s > 0$ and the function g of § 9.1.

We use integrals over phase space $\mathbb{R}^2$ of continuous functions bounded by $Ce^{-cg(z)}$ for constants C and $c > 0$, and over $\mathbb{R}^4$ of continuous functions bounded by $Ce^{-c(g(v)+g(z))}$. These are limits of integrals over squares, as in calculus. On a square, a continuous function can be integrated in either order, and the bound makes the part outside a large square small. So the integrals over $\mathbb{R}^4$ can be taken in either order, and a shift of the variables or the reflection $z \mapsto -z$ does not change any of them. For a Weyl pair, and vectors φ and ψ, the function $z \mapsto \langle \varphi, W'(z)\psi\rangle$ is continuous by Proposition 9.4, and bounded by $\|\varphi\|\,\|\psi\|$. Since $\int e^{-q^2/4s^2}\,\mathrm{d}q = 2s\sqrt{\pi}$ and $\int e^{-s^2p^2/4\hbar^2}\,\mathrm{d}p = 2\hbar\sqrt{\pi}/s$, by Lemma 8.5 at $k = 0$,

$$\int e^{-g(z)/4}\,\mathrm{d}z = 4\pi\hbar. \tag{9.2}$$

An operator can be given by its matrix elements. The key fact is the Riesz–Fréchet theorem, which *Linear Algebra*, Thm 13.40, cites. We prove what we need from Lemma 6.3.

Lemma 9.6 (An operator from its matrix elements)**.** *Let $B(\varphi, \psi)$ be a number for all $\varphi, \psi \in \mathcal{H}$, linear in ψ and antilinear in φ, with $|B(\varphi, \psi)| \le C\,\|\varphi\|\,\|\psi\|$. Then there is exactly one bounded operator A with $\langle \varphi, A\psi\rangle = B(\varphi, \psi)$ for all φ and ψ.*

Proof. Fix ψ, and let $\ell(\varphi) = \overline{B(\varphi, \psi)}$, which is linear in φ. Its kernel N is a subspace, and closed, since $|\ell(\varphi) - \ell(\varphi_n)| \le C\,\|\varphi - \varphi_n\|\,\|\psi\|$. If $\ell = 0$, put $A\psi = 0$. Otherwise $N \neq \mathcal{H}$, and by Lemma 6.3 there is a unit vector e orthogonal to N. For every φ, the vector $\ell(\varphi)e - \ell(e)\varphi$ lies in N, so it is orthogonal to e, and $\ell(\varphi) = \ell(e)\,\langle e, \varphi\rangle$. Put $A\psi = \overline{\ell(e)}\,e$. Then $\langle \varphi, A\psi\rangle = \overline{\langle A\psi, \varphi\rangle} = \overline{\ell(\varphi)} = B(\varphi, \psi)$. A vector orthogonal to every φ is 0, so $A\psi$ is the only such vector, that is, the only vector ξ with $\langle \varphi, \xi\rangle = B(\varphi, \psi)$ for every φ. Hence A is linear, since B is, and $\|A\psi\|^2 = B(A\psi, \psi) \le C\,\|A\psi\|\,\|\psi\|$ gives $\|A\psi\| \le C\,\|\psi\|$. ∎

Definition 9.7 (Von Neumann's operator)**.** For a Weyl pair on $\mathcal{H}$, the operator E_G is the bounded operator with

$$\langle \varphi, E_G\psi\rangle = \frac{1}{2\pi\hbar}\int e^{-g(z)/4}\,\langle \varphi, W'(z)\psi\rangle\,\mathrm{d}z \quad \text{for all } \varphi, \psi \in \mathcal{H}.$$

The right side is linear in ψ and antilinear in φ, and by (9.2) its modulus is at most $2\,\|\varphi\|\,\|\psi\|$. So Lemma 9.6 gives E_G. The letter G is for the Gaussian. On the line, by (9.1), the weight is $\langle h_0, W(z)h_0\rangle$, and § 9.4 shows that E_G is then $|h_0\rangle\langle h_0|$.

Lemma 9.8 (Von Neumann's projection)**.** *The operator E_G is self-adjoint, and for every point w of phase space*

$$E_G W'(w) E_G = e^{-g(w)/4}\, E_G. \tag{9.3}$$

In particular $E_G^2 = E_G$, so E_G is an orthogonal projection.

Proof. First, by the definition, and since the weight is real,

$$\langle E_G\varphi, \psi\rangle = \overline{\langle \psi, E_G\varphi\rangle} = \frac{1}{2\pi\hbar}\int e^{-g(z)/4}\,\langle W'(z)\varphi, \psi\rangle\,\mathrm{d}z.$$

Since $\langle W'(z)\varphi,\psi\rangle = \langle \varphi, W'(-z)\psi\rangle$ and $g(-z) = g(z)$, the reflection $z \mapsto -z$ turns this into $\langle \varphi, E_G\psi\rangle$. So E_G is self-adjoint.

Next, let $\varphi, \psi \in \mathcal{H}$. By the definition applied twice, and $W'(v)^\dagger = W'(-v)$ and $W'(w)^\dagger = W'(-w)$,

$$\langle \varphi, E_G W'(w) E_G \psi\rangle = \frac{1}{(2\pi\hbar)^2}\iint e^{-(g(v)+g(z))/4}\,\langle \varphi, W'(v)W'(w)W'(z)\psi\rangle\,\mathrm{d}v\,\mathrm{d}z.$$

By the product law, used twice, $W'(v)W'(w)W'(z)$ is $e^{\mathrm{i}[\sigma(w,z)+\sigma(v,w+z)]/2\hbar}W'(v+w+z)$. Substitute $z = u - w - v$ for fixed v. Since $\sigma(z,w)$ is linear in z and in w, and $\sigma(w,w) = 0$, the phase becomes $\sigma(w,u) + \sigma(v,u+w)$. The new integrand is still of our kind in (v,u). Indeed, $u = v + w + z$ gives $g(u) \le 3(g(v) + g(z) + g(w))$, so $g(v) + g(z) \ge (g(v) + g(u))/6 - g(w)/2$. Integrating over v first,

$$\langle \varphi, E_G W'(w) E_G \psi\rangle = \frac{1}{2\pi\hbar}\int K(u)\,\langle \varphi, W'(u)\psi\rangle\,\mathrm{d}u,$$

where, with $c = u - w$,

$$K(u) = \frac{e^{\mathrm{i}\sigma(w,u)/2\hbar}}{2\pi\hbar}\int e^{-(g(v)+g(c-v))/4}\,e^{\mathrm{i}\sigma(v,u+w)/2\hbar}\,\mathrm{d}v.$$

Expanding the squares gives $g(v) + g(c-v) = 2g(v - c/2) + g(c)/2$. Put $v = c/2 + y$. Then $\sigma(c/2, u+w) = \frac{1}{2}\sigma(u-w, u+w) = \sigma(u,w)$, whose phase cancels $e^{\mathrm{i}\sigma(w,u)/2\hbar}$. With $d = u + w = (d_q, d_p)$ and $y = (y_q, y_p)$, the phase $\sigma(y,d) = d_q y_p - y_q d_p$ splits, and Lemma 8.5, applied to each variable with $b = \sqrt{2}\,s$ and with $b = \sqrt{2}\,\hbar/s$, gives

$$\int e^{-g(y)/2} e^{\mathrm{i}\sigma(y,d)/2\hbar}\,\mathrm{d}y = 2\pi\hbar\, e^{-g(d)/8}.$$

So $K(u) = e^{-g(u-w)/8 - g(u+w)/8} = e^{-g(u)/4}e^{-g(w)/4}$, and the right side is $e^{-g(w)/4}\,\langle \varphi, E_G\psi\rangle$. This is (9.3). With $w = 0$, it reads $E_G^2 = E_G$. Then $E_G\psi$ is orthogonal to $\psi - E_G\psi$ for every ψ, since $\langle E_G\psi, \psi - E_G\psi\rangle = \langle \psi, E_G\psi - E_G^2\psi\rangle = 0$. ∎

Hall, Lemma 14.11, proves the same identity by checking it on $L^2(\mathbb{R})$ first. A projection may be 0, and $E_G = 0$ would make the lemma empty. It is not.

Lemma 9.9 (The projection is not zero)**.** *For every Weyl pair, $E_G \neq 0$.*

Proof. Let ψ be a unit vector, and $f(u) = \langle \psi, W'(u)\psi \rangle$. Then f is continuous, $|f| \le 1$ and $f(0) = 1$. For $\epsilon > 0$, consider

$$N_\epsilon = \frac{1}{2\pi\hbar} \int e^{-\epsilon g(z)} \, \langle W'(z)\psi, E_G W'(z)\psi \rangle \, \mathrm{d}z.$$

If $E_G = 0$, then $N_\epsilon = 0$. By the product law, $W'(-z)W'(u)W'(z) = e^{\mathrm{i}\sigma(u,z)/\hbar} W'(u)$. So by the definition of E_G,

$$N_\epsilon = \frac{1}{(2\pi\hbar)^2} \iint e^{-\epsilon g(z)} e^{-g(u)/4} e^{\mathrm{i}\sigma(u,z)/\hbar} f(u) \, \mathrm{d}u \, \mathrm{d}z.$$

For fixed u, Lemma 8.5 in each variable gives $\int e^{-\epsilon g(z)} e^{\mathrm{i}\sigma(u,z)/\hbar} \mathrm{d}z = (\pi\hbar/\epsilon) e^{-g(u)/4\epsilon}$. So $N_\epsilon = \int \gamma_\epsilon f \, \mathrm{d}u$, with the positive weight

$$\gamma_\epsilon(u) = \frac{1}{4\pi\hbar\epsilon} \, e^{-(1+\epsilon)g(u)/4\epsilon}, \qquad \int \gamma_\epsilon \, \mathrm{d}u = \frac{1}{1+\epsilon}, \qquad \int \gamma_\epsilon g \, \mathrm{d}u = \frac{4\epsilon}{(1+\epsilon)^2},$$

since the integrals behind (9.2) give $\int e^{-\lambda g(u)/4} \mathrm{d}u = 4\pi\hbar/\lambda$ for every $\lambda > 0$, and, by differentiating in λ, $\int g \, e^{-\lambda g(u)/4} \mathrm{d}u = 16\pi\hbar/\lambda^2$. As $\epsilon \to 0$, the weight concentrates at $u = 0$. Let $\eta > 0$, and choose $\delta > 0$ with $|f(u) - 1| < \eta$ when $g(u) \le \delta$. Since $|f - 1| \le 2$, this gives $|f(u) - 1| \le \eta + 2g(u)/\delta$ for every u. So

$$|N_\epsilon - 1| \le \int \gamma_\epsilon \, |f - 1| \, \mathrm{d}u + \frac{\epsilon}{1+\epsilon} \le \eta + \frac{8\epsilon}{\delta} + \epsilon,$$

which is less than 2η for small ϵ. So $N_\epsilon \to 1$, and $E_G \neq 0$. ∎

The integral N_ϵ tends to 1 for every unit vector, whereas $E_G = 0$ would make every N_ϵ vanish. Prugovečki, Lemma IV.6.3, and Hall, Prop. 14.10, prove $E_G \neq 0$ with Fourier series and the Fourier transform of $\mathbb{R}^2$.

Self-check 9.3

1. Use (9.3) to show that $\langle \chi, W'(z)\chi \rangle = e^{-g(z)/4}$ for every unit vector χ in the range of E_G.

9.4 Proof of the Stone–von Neumann theorem

Proof of Theorem 9.5.

1. *A vector.* By Lemma 9.9, the range of E_G contains a unit vector χ. By (9.3), $E_G\chi = \chi$ and the self-adjointness of E_G, $\langle \chi, W'(z)\chi \rangle = \langle \chi, E_G W'(z) E_G \chi \rangle = e^{-g(z)/4}$. By the product law, $W'(z)^\dagger W'(w) = W'(-z)W'(w) = e^{\mathrm{i}\sigma(w,z)/2\hbar} W'(w - z)$, so

$$\langle W'(z)\chi, W'(w)\chi \rangle = e^{\mathrm{i}\sigma(w,z)/2\hbar} \, e^{-g(w-z)/4} \quad \text{for all } z \text{ and } w. \tag{9.4}$$

2. *Its translates span $\mathcal{H}$.* Let $\mathcal{K}$ be the closure of the finite combinations of the vectors $W'(z)\chi$. By the product law, each $W'(w)$ maps $W'(z)\chi$ to a multiple of $W'(w+z)\chi$. So it maps the combinations into themselves, and, being bounded, it maps $\mathcal{K}$ into $\mathcal{K}$. In particular $U'(a) = W'(a, 0)$ and $V'(b) = W'(0, b)$ do. Since $\chi \in \mathcal{K}$ and the pair is irreducible, $\mathcal{K} = \mathcal{H}$.
3. *The same on the line.* By (9.1) and the same product law, the vectors $W(z)h_0$ satisfy (9.4) with h_0 in place of χ. Their closed span contains every $W(q, 0)h_0 = h_0(x - q)$. By Proposition 8.11, only 0 is orthogonal to all of these, so by Lemma 6.3 the closed span is $L^2(\mathbb{R})$.

4. *The unitary.* On a finite combination, put $T\big(\sum_j c_j W'(z_j)\chi\big) = \sum_j c_j W(z_j)h_0$. By the first and the third step, both sides have the same squared norm

$$\sum_{j,k} \overline{c_j} c_k \, e^{\mathrm{i}\sigma(z_k,z_j)/2\hbar} e^{-g(z_k-z_j)/4}.$$

So two combinations that are equal have equal images, since their difference has norm 0, and T is a well-defined linear map that keeps norms. If ξ_n are combinations with $\xi_n \to \xi$, the images form a Cauchy sequence, and $T\xi$ is their limit. The limit does not depend on the sequence, since $\|T\xi_n - T\xi_n'\| = \|\xi_n - \xi_n'\|$. This defines T on $\mathcal{H}$, by the second step, and T keeps norms. Its range is complete, so closed, and it contains every $W(z)h_0$, so by the third step it is $L^2(\mathbb{R})$. A linear map that keeps norms keeps inner products, by the polarisation identity. So $\langle T\xi, T\zeta\rangle = \langle \xi, \zeta\rangle$, and since T is onto, its inverse is its adjoint. So T is unitary.

5. *It carries one pair to the other.* By the product law,

$$TW'(w)W'(z)\chi = e^{\mathrm{i}\sigma(w,z)/2\hbar}\, TW'(w+z)\chi = e^{\mathrm{i}\sigma(w,z)/2\hbar}\, W(w+z)h_0,$$

which is $W(w)W(z)h_0 = W(w)TW'(z)\chi$. So $TW'(w) = W(w)T$ on the combinations, and, since both sides are bounded, on $\mathcal{H}$. With $w = (a,0)$ and $w = (0,b)$, this is the claim. ∎

The range of E_G is the one-dimensional subspace spanned by χ. Indeed, let χ_1 be in the range and orthogonal to χ. Then, by (9.3),

$$\langle \chi_1, W'(z)\chi\rangle = \langle \chi_1, E_G W'(z) E_G \chi\rangle = e^{-g(z)/4}\,\langle \chi_1, \chi\rangle = 0 \quad \text{for every } z.$$

By the second step, χ_1 is orthogonal to all of $\mathcal{H}$, so $\chi_1 = 0$. So $E_G = |\chi\rangle\langle\chi|$, and T maps χ to h_0.

The Schrödinger pair is itself irreducible.

Proposition 9.10 (The Schrödinger pair is irreducible)**.** *The only closed subspaces of $L^2(\mathbb{R})$ that every $U(a)$ and every $V(b)$ map into themselves are $\{0\}$ and $L^2(\mathbb{R})$.*

Proof. Let $\mathcal{K} \neq \{0\}$ be such a subspace. Since $U(a)^\dagger = U(-a)$ and $V(b)^\dagger = V(-b)$, the restrictions to $\mathcal{K}$ are a Weyl pair on $\mathcal{K}$, with the restrictions of the $W(z)$ as its Weyl operators. By Lemmas 9.8 and 9.9 for that pair, there is a unit vector $\xi \in \mathcal{K}$ with $\langle \xi, W(z)\xi\rangle = e^{-g(z)/4}$ for every z, as in the first step of the proof above. By (9.1), h_0 has the same property. Let E_G now be the operator of the whole pair on $L^2(\mathbb{R})$. For a unit vector η with this property, $\langle \eta, E_G\eta\rangle = (2\pi\hbar)^{-1}\int e^{-g(z)/2}\mathrm{d}z = 1$, and since E_G is a projection, $\|\eta - E_G\eta\|^2 = 1 - \langle \eta, E_G\eta\rangle = 0$. So $E_G\xi = \xi$ and $E_G h_0 = h_0$. Write $\xi = ch_0 + \xi_1$ with $c = \langle h_0, \xi\rangle$. Then $E_G\xi_1 = \xi_1$ and $\langle h_0, \xi_1\rangle = 0$, and by (9.3)

$$\langle W(z)h_0, \xi_1\rangle = \langle h_0, E_G W(-z) E_G \xi_1\rangle = e^{-g(z)/4}\,\langle h_0, \xi_1\rangle = 0 \quad \text{for every } z.$$

By the third step of the proof, $\xi_1 = 0$. So $c \neq 0$, and $h_0 = \xi/c$ lies in $\mathcal{K}$. Then so does every $W(z)h_0$, and $\mathcal{K} = L^2(\mathbb{R})$ by the third step again. ∎

This is Hall, Prop. 14.7. The proof also shows that the range of the line's E_G is spanned by h_0, so there $E_G = |h_0\rangle\langle h_0|$.

A particle with spin $\frac{1}{2}$ is not of this kind. Its states are pairs (ψ_0, ψ_1) of functions, the groups act on both, and the pairs $(\psi_0, 0)$ form a closed subspace that both groups keep.

Self-check 9.4

1. Let $\mathcal{H} = L^2(\mathbb{R}) \oplus L^2(\mathbb{R})$, the pairs (ψ_0, ψ_1), with $U'(a)$ and $V'(b)$ acting on both functions as $U(a)$ and $V(b)$ do. What is E_G? What is the dimension of its range?

9.5 What the theorem decides

The theorem answers the question of this lecture. A particle on the line without further degrees of freedom has the state space $L^2(\mathbb{R})$, with Q and P on their domains of Theorem 4.6. This holds as soon as its translations in position and in momentum are strongly continuous, obey Weyl's relation and act irreducibly. No other realisation exists, up to a unitary map.

Besides the continuity of the two groups, it assumes three things. Weyl's relation, not the commutator on a dense domain, as the box shows. Irreducibility, which excludes a spin. One value of $\hbar$. It decides nothing about the Hamiltonian. The free particle and the oscillator share the pair and differ in their energies.

The canonical pair

States: for a particle on the line with no further degree of freedom, every irreducible Weyl pair is the line's, up to a unitary map (Theorem 9.5). So its state space is $L^2(\mathbb{R})$. *Observables:* the generators of the two groups are $P/\hbar$ and $-Q/\hbar$, with the domains of P and Q, and $U(a)V(b) = e^{-\mathrm{i}ab/\hbar}V(b)U(a)$ (Proposition 9.2). The relation $QP - PQ = \mathrm{i}\hbar$ on a dense domain does not fix them: in the box it holds and Weyl's relation fails. *Evolution:* the theorem says nothing about the Hamiltonian. *Measurement:* for an irreducible pair, von Neumann's projection E_G, built from the Weyl operators alone, is the projection onto one vector, which the unitary map carries to the oscillator's ground state h_0 (§ 9.4, after the proof of Theorem 9.5).

Summary.

- *Momentum generates translations*, $U(a) = e^{-\mathrm{i}aP/\hbar}$ (Proposition 9.1), and $V(b) = e^{\mathrm{i}bQ/\hbar}$; *Weyl's relation* $U(a)V(b) = e^{-\mathrm{i}ab/\hbar}V(b)U(a)$ (Proposition 9.2).
- *The product law* $W'(z)W'(w) = e^{\mathrm{i}\sigma(z,w)/2\hbar}W'(z+w)$ for every Weyl pair (Proposition 9.4).
- *Von Neumann's projection*: E_G is an orthogonal projection with $E_G W'(w) E_G = e^{-g(w)/4}E_G$, and $E_G \neq 0$ (Lemmas 9.8 and 9.9).
- *Stone–von Neumann*: every irreducible Weyl pair is unitarily equivalent to the Schrödinger pair (Theorem 9.5), which is irreducible (Proposition 9.10).

Lecture 10 adds a potential to the free energy: the pair stays, and the Hamiltonian changes.

Self-check answers.

9.1 1. Substituting $y = x - a$, $\int x\,|\psi(x-a)|^2\,\mathrm{d}x = \int (y+a)\,|\psi(y)|^2\,\mathrm{d}y = \langle Q\rangle + a$, since $\|\psi\| = 1$.

9.2 1. On the eigenvector $e_{\theta,k}$, the group multiplies by $e^{-\mathrm{i}L(\theta+2\pi k)/L} = e^{-\mathrm{i}\theta}$, the same for every k. Two bounded operators that agree on a basis agree everywhere.

9.3 1. Since $E_G\chi = \chi$ and E_G is self-adjoint, $\langle \chi, W'(z)\chi\rangle = \langle E_G\chi, W'(z)E_G\chi\rangle = \langle \chi, E_G W'(z) E_G\chi\rangle = e^{-g(z)/4}\,\|\chi\|^2$.

9.4 1. The integrand acts on each function, so $E_G(\psi_0,\psi_1) = (\langle h_0,\psi_0\rangle\, h_0, \langle h_0,\psi_1\rangle\, h_0)$. Its range is spanned by $(h_0, 0)$ and $(0, h_0)$: dimension 2.

References

Hall, *Quantum Theory for Mathematicians*, Ex. 14.5 (p. 285), Def. 14.6, Prop. 14.7 and Thm 14.8 (p. 286), Prop. 14.10 (pp. 288–289), Lemma 14.11 (pp. 289–290). Prugovečki, *Quantum Mechanics in Hilbert Space*, Thm IV.6.4 (p. 342), §IV.6.5 (pp. 342–347), Lemma IV.6.3 (p. 344). *Linear Algebra for Physics*, Thm 13.40. *Quantum Mechanics I*, Thm 9.2 and Exercise Sheet 9.